UNIVERSITY OF HERTFORDSHIRE

School of Physics, Engineering and Computer Science

Title: Analyzing the Impact of Network Depth in Multilayer Perceptrons

Student Name: Sanjana

Student ID : 24137661

1. Introduction

Machine Learning is a key area of modern computer science that enables systems to learn patterns from data and make intelligent predictions. Among various ML techniques, neural networks—especially Multilayer Perceptrons (MLPs)—are widely used for solving complex classification problems.

An MLP is a feedforward neural network composed of multiple layers of neurons, including input, hidden, and output layers. One of the most important design choices in an MLP is its **depth**, which refers to the number of hidden layers.

This tutorial focuses on analyzing how increasing the depth of an MLP affects its performance, particularly in terms of accuracy and overfitting.

2. Objective

The objective of this tutorial is to investigate the effect of network depth on model performance. Specifically, we aim to:

- Compare training and testing accuracy across different depths
- Understand the concept of overfitting in deep networks
- Provide practical insights into selecting optimal model complexity

3. Methodology

3.1 Dataset

The Breast Cancer dataset from the Scikit-learn library is used for this experiment. It consists of 569 samples and 30 numerical features, making it suitable for binary classification tasks.

3.2 Preprocessing

The dataset is split into training and testing sets using an 80:20 ratio. Feature scaling is applied using StandardScaler to ensure all features are on a similar scale, which improves neural network performance.

3.3 Model Configuration

Four Multilayer Perceptron models are trained with varying depths:

- 1 hidden layer
- 2 hidden layers
- 3 hidden layers
- 4 hidden layers

All models are trained using the same parameters to ensure a fair comparison.

4. Results and Analysis

The results show that training accuracy increases as the number of layers increases. However, test accuracy improves initially and then decreases for deeper models. This indicates that deeper networks tend to overfit the training data, reducing their ability to generalize to unseen data.

Figure 1 shows the effect of depth on performance.

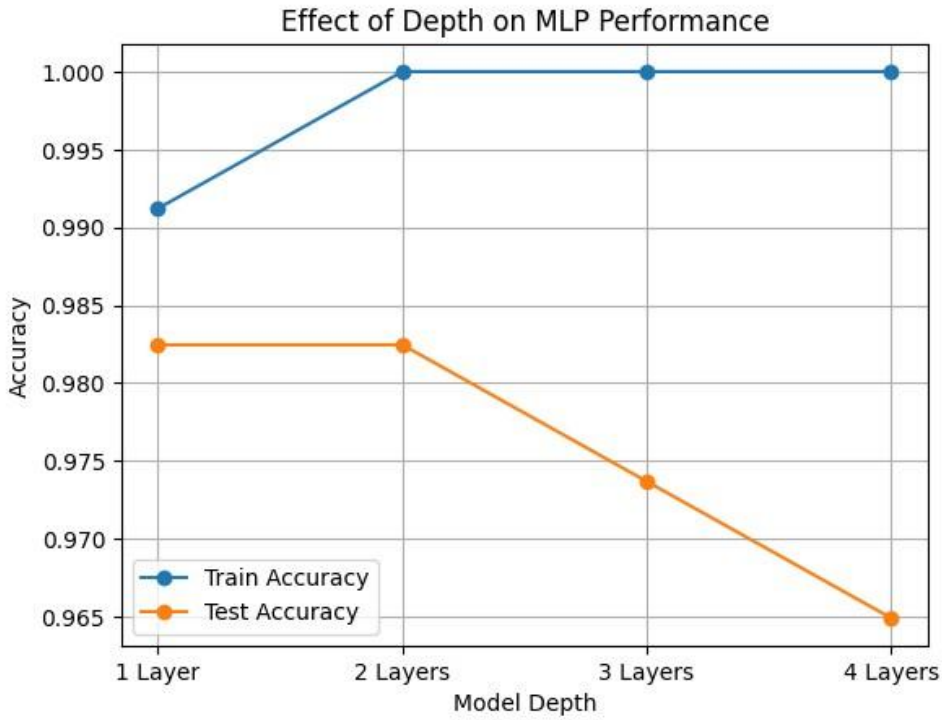

Figure 1: Effect of Depth on MLP Performance

The confusion matrix demonstrates strong classification performance, with most predictions correctly classified. Only a small number of misclassifications are observed, indicating that the model is effective.

Figure 2 shows the confusion matrix of the best model.

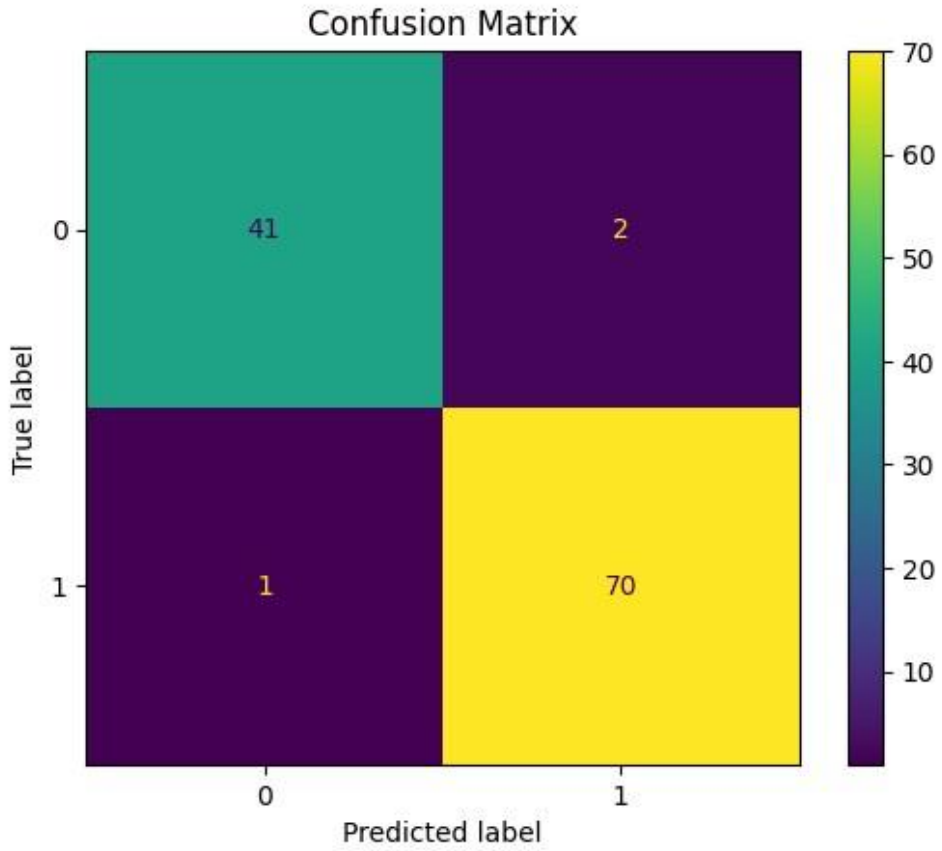

Figure 2: Confusion Matrix of MLP Model

5. Discussion

The experiment highlights the trade-off between model complexity and generalization. While deeper neural networks have greater learning capacity, they are more likely to overfit the training data.

This emphasizes the importance of selecting an optimal depth rather than simply increasing complexity. Techniques such as regularization, dropout, and cross-validation can help mitigate overfitting in deep neural networks.

6. Conclusion

This tutorial demonstrated how the depth of a Multilayer Perceptron impacts its performance. The key findings are:

- Increasing depth improves training accuracy
- Test accuracy may decrease due to overfitting

- Optimal depth is essential for achieving good generalization

Understanding these trade-offs is crucial for designing effective machine learning models.

6. Personal Reflection

Through this tutorial, I gained practical insight into how neural network architecture influences performance. By experimenting with different depths, I understood the importance of balancing model complexity and generalization.

This experience improved my understanding of machine learning concepts and strengthened my ability to design and evaluate models effectively.